Edge-based energy efficient loT driven smart waste management system

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Municipal waste management in Sri Lanka continues to face irregular collection schedules, localized bin overflow, weak visibility into field conditions, and inefficient coordination between households and collection operators. IoT-based waste management systems improve monitoring by attaching sensors to bins, but many rely on frequent transmission of raw sensor readings to central servers. This increases communication cost, bandwidth, drains device batteries, and limits scalability when many bins are deployed. Existing systems also place little emphasis on adaptive edge intelligence, predictive local decisions, or integrated user-facing services that support both operators and households.

This proposal presents an edge-based energy efficient IoT smart waste management system. Low power smart-bin nodes built around ESP32 class controllers will sense fill level, weight, gas/odor, and battery state. A Raspberry Pi-class edge gateway will manage a small cluster of bins, apply rule-based filtering, estimate short term fill trends, and adapt outgoing data according to battery condition and network quality. Only operationally valuable information will be forwarded to cloud services, the collection and monitoring web application, and a mobile application for home users. The study will evaluate whether the proposed architecture can reduce transmissions and improve energy efficiency while preserving timely alerts, route planning usefulness, and user service quality.

The project will use embedded IoT hardware, edge computing, RESTful backend services, a web dashboard, and a mobile client. Expected outcomes include a working prototype, quantitative evaluation of communication and energy savings, and a practical architecture suitable for urban waste management operations. The solution has social value through cleaner neighborhoods and better service responsiveness, and commercial value through reduced operational cost and improved scalability for municipal and private waste operators.

Keywords: IoT, edge computing, smart waste management, energy efficiency, predictive monitoring.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ESP32	Low-power microcontroller family used for IoT nodes
GPS	Global Positioning System
IoT	Internet of Things
LKR	Sri Lankan Rupees
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience

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1. INTRODUCTION

1.1 Background and Context

Municipal solid waste management remains a persistent operational and environmental problem in Sri Lanka. Rapid urbanization, population growth, inconsistent source segregation, and uneven collection capacity create recurring issues such as overflowing bins, illegal dumping, odor, and delayed response to local incidents. Households, local authorities, collection contractors, and surrounding communities are directly affected by these inefficiencies. The problem is not only a public-service issue but also a public-health and sustainability issue because unmanaged waste contributes to environmental pollution, poor sanitation, and avoidable operational cost [1], [2].

IoT-based waste-management systems have been proposed to improve visibility by instrumenting bins with sensors and connecting them to monitoring platforms [1] [3]. In theory, real-time sensing can support better route planning and faster alerts. In practice, however, many systems stream raw data frequently to centralized services. This design choice creates a different problem: a city-scale deployment can generate many low-value transmissions, especially when bins are stable for long periods. For battery-powered or low-power nodes, the communication burden becomes a major source of energy use. At the same time, centralized processing introduces avoidable bandwidth usage and dependence on continuous internet availability.

Edge computing offers a more suitable architecture for this scenario because part of the intelligence can be placed near the data source rather than only in the cloud [4] [5]. In the proposed solution, smart-bin nodes collect sensing data, while a nearby edge gateway coordinates a cluster of bins, filters redundant updates, predicts short-term bin conditions, and adapts its communication behavior using battery and network state. The system also includes a collection and monitoring web application for operators and a mobile application for home users. This combination aims to improve operational efficiency, user responsiveness, and scalability while keeping device-side complexity practical.

The proposed project aligns with SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production) by targeting cleaner cities, better resource use, and improved waste operations. It also aligns with the Software Systems & Technologies (SST) research cluster because the main contribution is the design and evaluation of an edge-based IoT system architecture, adaptive algorithms, and integrated software services.

1.2 Literature Review

Study	Focus	Strengths	Limitations	Relevance
[1], [6]	IoT smart bins and wireless monitoring	Demonstrate feasibility of connected waste systems	Mostly centralized communication; limited energy strategy	Baseline waste-sensing architecture
[7]	Sensor-driven smart waste operations	Useful for municipal agent routing and field awareness	Limited emphasis on adaptive data reduction at the edge	Operational motivation
[3]	Wireless-sensor architecture for waste management	Highlights distributed sensing design	Predates modern edge-computing patterns	Early system design reference
[8], [9]	LoRa / IoT / intelligent waste systems	Improved field connectivity and classification	Focus often on communication technology or analytics, not adaptive edge gateways	Shows advanced variants
[4], [10], [5]	Edge-computing architectures	Explain why local processing reduces latency and cloud dependence	Not specific to waste management	Architectural foundation
[11], [12]	Smart-city / edge-IoT surveys	Broader view of scalable and energy-aware urban IoT	Do not define a concrete waste-management stack with both operator and household interfaces	Supports design decisions

Table 1

Existing research demonstrates that smart waste bins can generate useful operational data, and that route planning can benefit from timely condition updates [1], [9]. Several systems employ fill-level sensing, wireless networks, or dashboard monitoring. However, much of the literature treats each bin as a direct sender to a remote server, leaving limited discussion of how communication should be reduced when sensor states remain stable or when local context suggests that forwarding every reading adds little value.

Parallel literature on edge computing argues that pushing computation closer to devices can reduce latency, lower backhaul traffic, and improve responsiveness in IoT systems [4], [5]. These benefits are highly relevant to waste management, where many sensor updates are repetitive, while a small subset of events such as overflow risk or gas/odor anomalies - require prompt action. Edge computing also creates an opportunity to combine rule-based decisions and predictive estimation without overloading small bin controllers.

Commercial and research prototypes also tend to emphasize either operator dashboards or sensing hardware, but fewer solutions provide a coherent design that includes smart-bin nodes, a cluster edge gateway, a collection web platform, and a home-user mobile application within one energy-aware architecture. This gap motivates the present proposal

1.3 Research Gap

The key research gap is the lack of an integrated, energy aware edge architecture for smart waste management that explicitly decides what data should be forwarded, when it should be forwarded, and at what level of detail. Current waste-monitoring solutions generally prove that connected bins are possible, but they do not adequately address how to reduce communication overhead in resource constrained deployments while still supporting operational monitoring and user-facing services.

The proposed project addresses this gap by introducing a hierarchical design in which low-power IoT bin nodes are paired with a cluster edge gateway. The gateway performs rule-based filtering, short term predictive estimation, and adaptive output control using battery and network status. Unlike a purely cloud-centric design, the proposed system aims to preserve useful alerts and route-planning signals while reducing unnecessary transmissions. The inclusion of both a collection and monitoring web application and a home-user mobile application further extends the novelty from hardware sensing alone to a practical service ecosystem.

The creativity of the project lies not in adding more raw sensing, but in deciding how sensing data should be transformed into operationally useful outputs at the edge. This makes the proposal both technically meaningful and commercially relevant

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate an edge-based energy-efficient IoT smart waste-management system that reduces unnecessary communication, supports predictive and adaptive edge-node functions, and improves operational monitoring and household-facing waste services

2.2 Specific Objectives

- Design and prototype a low power IoT smart-bin device using an ESP32 class controller and suitable sensors for fill level, weight, gas/odor, and battery monitoring.
- Develop an edge node that aggregates data from multiple bins, applies rule-based filtering, estimates short-term fill conditions, and adapts outgoing data according to battery level and network quality.
- Implement a collection and monitoring web application for operators to visualize bin states, manage alerts, review trends, and support route-planning decisions.
- Develop a mobile application for home users to receive collection schedules and alerts, report issues, and access disposal guidance.
- Evaluate the proposed architecture against a baseline raw-data transmission approach using metrics such as transmission count, estimated energy usage, latency of urgent alerts, and practical usefulness for waste-management decisions.

3. METHODOLOGY

This study will follow a design science and experimental evaluation approach. The design science element is used to build the proposed artifact - the smart-bin device, edge node, backend services, and user-facing applications while the experimental element is used to compare the edge-based approach against a simpler baseline that forwards raw data more frequently.

The IoT device layer will be implemented using an ESP32-class microcontroller with representative sensors. The edge layer will be implemented on a Raspberry Pi-class gateway. The gateway will use rule-based logic such as threshold checks, event classification, state transitions, and short-term trend estimation from recent readings. Adaptive output control will change transmission frequency or payload detail using network availability and battery thresholds. For example, urgent events will always be forwarded immediately, while routine stable states will be summarized or delayed.

The backend will expose RESTful APIs and a database to store device states, incidents, and user facing information. The collection and monitoring web application will visualize bin clusters, alerts, and route-support information. The mobile application will focus on household notifications, issue reporting, and service visibility. Suitable technologies include Python or Node.js for backend services, react for the web application, Flutter or React Native for the mobile application, and lightweight messaging such as MQTT or HTTP for device communication.

Data will be gathered from prototype sensors, simulated multi-bin streams, and scenario-based testing. Ethical considerations are limited because the system mainly handles operational sensor data and user-entered service reports rather than sensitive personal data, but privacy, secure authentication, and responsible data handling will still be applied. Validation will use metrics such as number of transmissions, estimated communication energy, alert latency, gateway decision accuracy, system uptime, and user-interface responsiveness.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system consists of five main components: (1) smart-bin IoT nodes, (2) a cluster edge gateway, (3) cloud backend services, (4) a collection and monitoring web application, and (5) a home user mobile application. Figure 1 presents the overall architecture.

Each smart-bin node senses fill level, weight, gas/odor, and battery state. Rather than sending every reading directly to the internet, nodes communicate with a nearby edge gateway. The edge gateway aggregates updates from a cluster of bins, applies rule-based filtering and short-term prediction, stores temporary state, and decides whether outgoing communication should be immediate, summarized, or deferred. This architecture enables local response even when internet conditions are weak.

The cloud backend persists data, handles authentication, and exposes APIs to the web and mobile clients. The operator-facing web application focuses on monitoring, analytics, and collection support. The mobile application targets households by providing service schedules, alerts, and issue reporting.

The architecture is scalable because multiple edge gateways can manage multiple bin clusters independently while synchronizing with shared cloud services. Security considerations include device authentication, role-based access to applications, encrypted transport, and controlled API exposure.

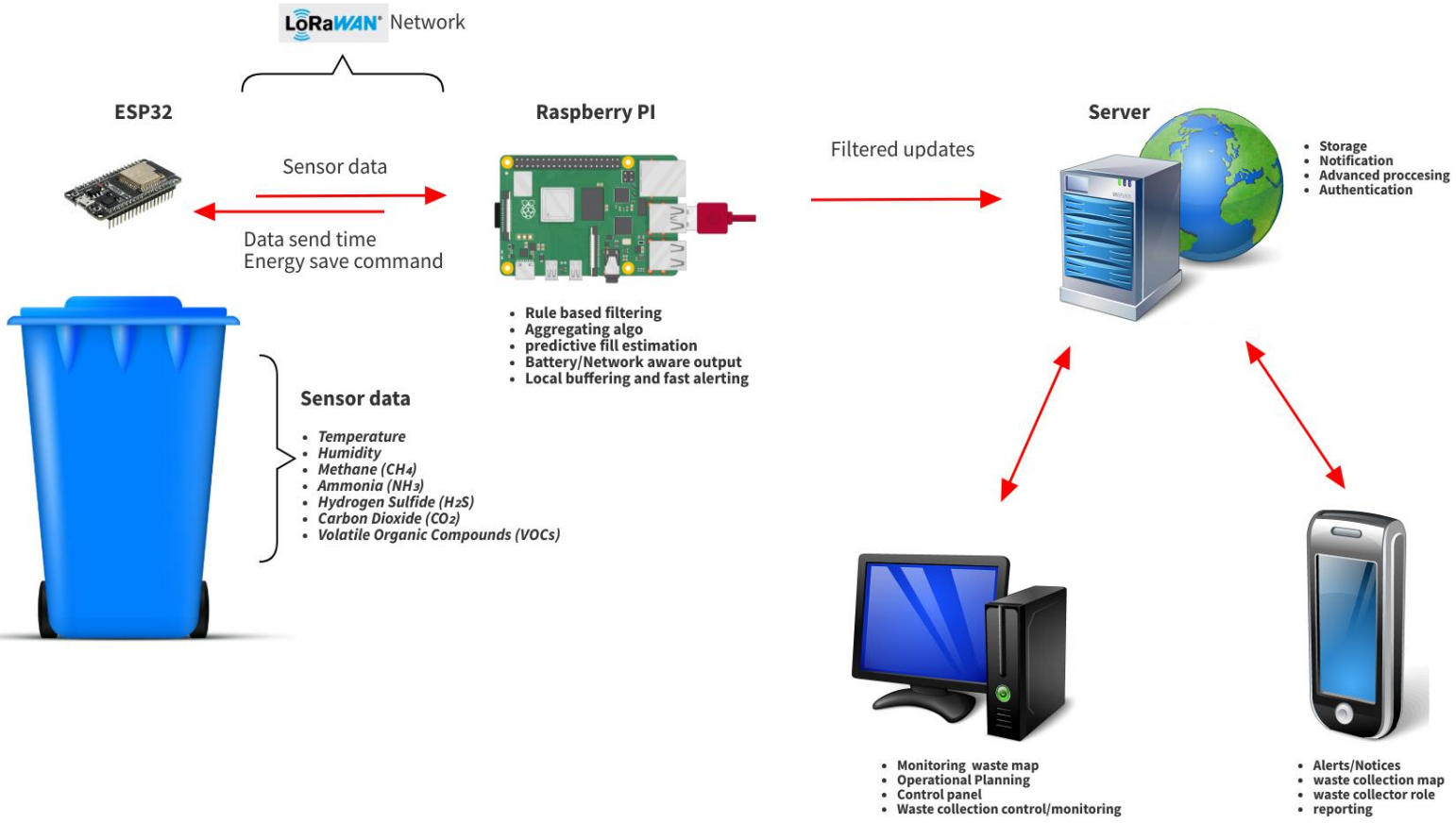


Figure 1.

5. USER REQUIREMENTS

Stakeholder	Needs	System Benefit
Waste-management administrators (Municipal / private)	• Timely alerts on bin status and overflow • Reduced operational and collection costs • Data-driven decision making • Monitoring of multiple bin clusters • Device health and fault visibility	Web dashboard, prediction support, cluster monitoring Reduced data traffic and improved efficiency Battery/network/device status monitoring
Household users	• Collection schedules, • Ability to report, feedback channel • cleaner surroundings	Mobile app alerts Reporting and feedback Service transparency
Collection staff (or volunteers)	• Clear collection routes and priorities • Real-time bin status updates • Reduced unnecessary trips	Priority-based collection alerts improved operational efficiency

Table 2

ID	Functional requirement	Priority
FR1	The IoT node shall capture fill-level, weight, gas/odor, and battery readings.	High
FR2	The IoT node shall send readings to the edge gateway using a lightweight communication method.	High
FR3	The edge gateway shall aggregate data from multiple bins.	High
FR4	The edge gateway shall apply rule-based filtering and event classification.	High
FR5	The edge gateway shall adapt output frequency/detail using battery and network state.	High
FR6	The web application shall display bin status, alerts, and historical trends.	High
FR7	The web application shall support route-planning decisions and zone-level monitoring.	Medium
FR8	The mobile application shall show schedules, alerts, and allow issue reporting.	High
FR9	The backend shall store device, alert, and user-report data securely.	High

Table 3

Category	Requirement	Rationale
Performance	Urgent alerts should reach the dashboard with low delay under normal connectivity.	Operational usefulness
Energy efficiency	The architecture should reduce unnecessary transmissions versus a baseline design.	Battery life and scalability
Reliability	The edge node should buffer data temporarily during internet disruption.	Continuity of service
Security	Application and API access should use authentication and encrypted transport.	Protect device and user data
Usability	Web and mobile interfaces should be simple for non-technical users.	Adoption and service value
Scalability	One gateway should support a small cluster of bins and allow expansion by replication.	Real deployment readiness

Table 4

6. COMMERCIALIZATION PLAN

The target market includes municipal councils, private waste-collection firms, residential complexes, universities, hospitals, and commercial estates that need better visibility into waste generation and collection. The primary customer persona is an operations manager responsible for collection efficiency, service reliability, and cost control. A secondary customer segment is property managers who want a cleaner environment with better resident communication.

The value proposition is a practical, low-cost smart waste-management platform that improves collection decisions while reducing communication waste and device energy consumption. Compared with a cloud-only model, the proposed system offers better field resilience, lower backhaul usage, and more intelligent data handling through the edge gateway. For end users, the mobile application improves transparency and reporting.

A realistic revenue model for commercialization is a hybrid of one-time hardware deployment charges and recurring software/service subscriptions. Hardware packages can be sold per smart-bin and edge-gateway cluster, while the software platform can be priced monthly for dashboard access, support, analytics, and maintenance. Additional revenue can come from managed services, custom deployments, and integration with municipal reporting systems.

Competitive advantage arises from combining low-cost hardware, adaptive edge intelligence, and two user facing channels , a web dashboard for operators and a mobile application for

households within one architecture. Intellectual property potential exists mainly in the system design, software implementation, deployment know-how, and future proprietary rule engines or optimization features.

7. BUDGET AND JUSTIFICATION

Item	Qty	Estimated Unit Cost (LKR)	Subtotal (LKR)	Justification
ESP32 development boards	2	3,500	7,000	Prototype smart-bin nodes
Ultrasonic / load / gas sensors	2	6,000	12,000	Representative sensing stack
Raspberry Pi-class edge gateway	1	28,000	28,000	Edge processing and local coordination
Power banks / batteries / cabling	1	8,000	8,000	Prototype power setup
Cloud / hosting / database	4	3,000	12,000	Backend deployment and testing
Domain / messaging / notification services	1	5,000	5,000	User notification support
Miscellaneous fabrication and transport	1	8,000	8,000	Assembly and testing overhead
Total			**80,000**	**Indicative prototype budget**

Table 5

Software licensing cost is expected to remain low by prioritizing open-source frameworks. Human effort is not monetized in this student proposal, but time investment is significant across hardware integration, gateway logic, frontend development, testing, and documentation.

The budget is intentionally prototype-oriented rather than deployment-oriented. A real municipal deployment would require rugged enclosures, larger-scale communications infrastructure, maintenance planning, and operational support

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Work package	Main outputs	Indicative weeks
1	Problem definition and literature review	Problem statement, survey, references	1-5
2	Requirements and architecture design	Requirements, diagrams, technology choices	2-4
3	IoT smart-bin prototype	ESP32 device logic and sensor integration	4-8
4	Edge gateway development	Rule engine, prediction, adaptive output control	6-10
5	Backend and database services	APIs, data storage, authentication	8-11
6	Collection and monitoring web app	Dashboard, alerts, analytics views	8-13
7	Mobile app for home users	Schedules, notifications, reporting	9-13
8	System integration	End-to-end communication and testing	11-14
9	Evaluation and analysis	Metrics, comparisons, result interpretation	13-16
10	Documentation and presentation	Proposal, progress deliverables, final report	14-18

Table 6

9. GANTT CHART

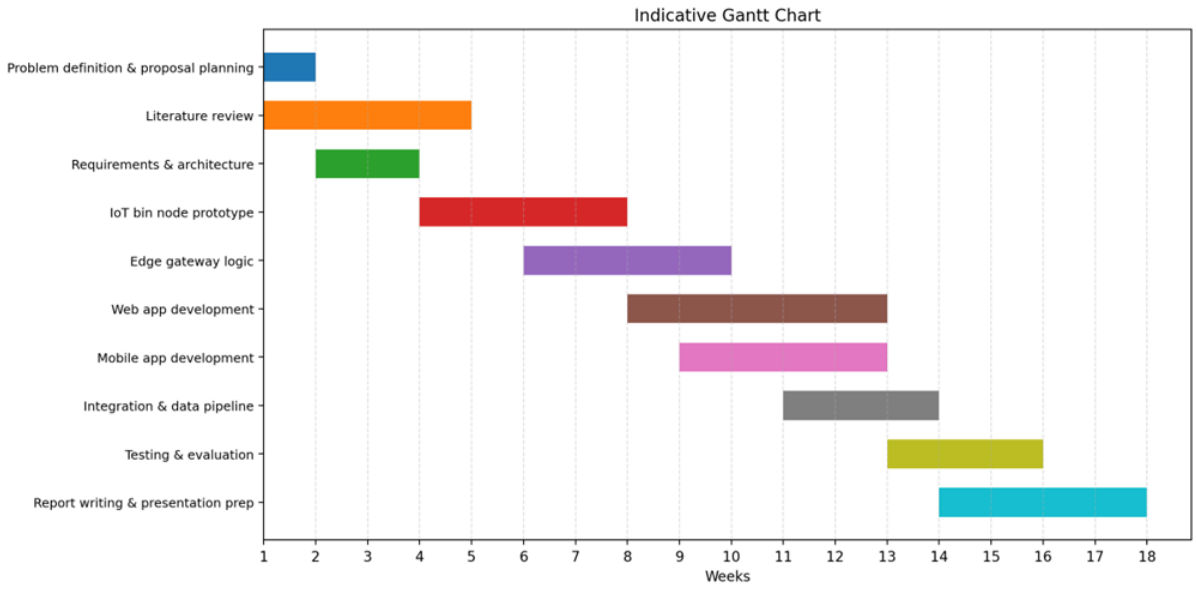


Figure 2.

REFERENCES LIST

[1] S. a. R. N. Thakker, "Smart and wireless waste management," in *2015 International Conference on Innovations in Information,Embedded and Communication Systems (ICIIECS)*, Coimbatore, India., 2015.

[2] Lal, Mervin and Dharmasiri, "Waste Management in Sri Lanka: Challenges and Opportunities," *Sri Lanka journal of social sciences,* vol. 09, no. may, pp. 75-85, 2020.

[3] S. a. M. D. a. A. E. a. D. B. G. a. P. M. a. G. M. a. P. M. Longhi, "Solid Waste Management Architecture Using Wireless Sensor Network Technology," in *New Technologies, Mobility and Security (NTMS)*, 2012.

[4] Shi, Weisong, Cao, Jie, Zhang, Quan, Li, Youhuizi and Xu, Lanyu, "Edge Computing: Vision and Challenges," *IEEE Internet of Things Journal,* vol. 3, p. 1, 2016/10/01.

[5] Premsankar, Gopika, Francesco, Mario, Taleb and Tarik, "Edge Computing for the Internet of Things: A Case Study," *IEEE Internet of Things Journal,* vol. pp, no. feb, pp. 1-1, 2018.

[6] Shyam, Gopal Kirshna, Manvi, Sunilkumar S, Bharti and Priyanka, "Smart waste management using Internet-of-Things (IoT)," in *International Conference on Computer and Communications Technologies (ICCCT)*, 2017.

[7] G. D. K. B. O. S. Omara A, "Trajectory-Assisted Municipal Agent Mobility: A Sensor-Driven Smart Waste Management System," *Journal of Sensor and Actuator Networks.,* vol. 7, no. 3, p. 29, 2018.

[8] T. a. S. I. M. a. M. N. a. B. M. H. a. A. H. a. I. M. R. a. C. M. a. R. H. a. I. M. Sheng, "An Internet of Things Based Smart Waste Management System Using LoRa And Tensorflow Deep Learning Model," vol. pp, pp. 1-1, 08 2020.

[9] V. a. M. V. a. R. E. a. V. B. Sankeerth, "Smart Waste Management System Using IoT," in *inventive Computation Technologies. ICICIT 2019*, 2020.

[10] Satyanarayanan and Mahadev, "The Emergence of Edge Computing," *Computer,* vol. 50, pp. 30-39, 2017/01/01.

[11] Tuwanut, P and Kraijak, S., "A survey on IoT architectures, protocols, applications, security, privacy, real-world implementation and future trends," *11th International Conference on Wireless Communications,* no. 11, pp. 6-6, 2015 jan.

[12] Ashkan Yousefpour, Caleb Fung, Tam Nguyen, Krishna Kadiyala, Fatemeh Jalali, Amirreza Niakanlahiji, Jian Kong and Jason P. Jue, "All one needs to know about fog computing and related edge computing paradigms: A complete survey," *Journal of Systems Architecture,* vol. 98, pp. 289-330, 2019.

10.APPENDICES

Appendix A. Initial interface notes

- The operator web application should include a map or zone-based summary, alert panel, device health status, and route-support indicators.
- The mobile application should remain lightweight and focus on collection schedules, incident reporting, and notifications rather than complex analytics.

Appendix B. Risk analysis summary

Risk	Impact	Mitigation	Residual level
Sensor inaccuracy	Incorrect alerts or trends	Calibration and scenario testing	Medium
Gateway failure	Loss of cluster coordination	Local buffering and restart strategy	Medium
Network instability	Delayed updates	Adaptive output and temporary edge storage	Medium
Scope creep	Delayed completion	Prioritize core architecture and evaluation	Low